

PERSPECTIVE

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4-MBC in Australian sunscreens as an unmonitored discharge pathway into Great Barrier Reef coastal waters

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Abstract

4-Methylbenzylidene camphor (4-MBC, also known as enzacamene), a chemical UV filter prohibited or never permitted across the EU, UK, China, ASEAN, US, and Japan, remains legal in Australian therapeutic sunscreens at concentrations up to 4% w/w. The European Scientific Committee on Consumer Safety concluded in 2022 that no safe concentration could be established, and the UK Scientific Advisory Group on Chemical Safety confirmed this in 2025. In January 2026 the Australian Therapeutic Goods Administration also concluded that a safe dose cannot be established, yet declined to remove the compound from the Permissible Ingredients Determination, citing supply-shortage grounds. This article examines a narrower question: whether 4-MBC is entering Great Barrier Reef catchment coastal waters through a direct, unmonitored wastewater pathway, and whether the concentrations reaching reef-adjacent waters are sufficient to compound bleaching in already-stressed coral systems. Two Queensland focus zones and one downstream transport corridor are examined where documented bleaching overlaps with confirmed wastewater treatment plant discharge points: the inshore reefs north of Cairns, Magnetic Island and Cleveland Bay, and the transport corridor from Townsville to the Palm Islands. Peer-reviewed evidence establishes 4-MBC as a coral toxicant in single-compound laboratory exposure, though at concentrations well above those measured in coastal waters. A twelve-compound UV filter mixture containing 4-MBC at 200 ng/L per compound has been shown to amplify thermal bleaching in reef corals. Whether 4-MBC occurs in Queensland reef-adjacent waters at concentrations relevant to either finding is unknown, because the discharge pathway has never been monitored for this class of contaminant. The case for action does not rest on causation of the 2024 bleaching event. It rests on the continuous discharge of a coral toxicant, banned by Australia's regulatory peers, through a pathway that has never been measured. This Perspective specifies the monitoring that would resolve the scientific question and identifies the existing Australian regulatory instrument through which alignment with the international scientific consensus could be achieved.

Keywords 4-MBC, Enzacamene, Sunscreen, Great Barrier Reef, Australia, Therapeutic goods administration



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1 Introduction

Chemical ultraviolet (UV) filters are an under-characterised class of marine contaminants. The compounds enter coastal waters continuously through bathing, wastewater discharge, and direct recreational wash-off. Their environmental fate has only recently begun to be measured at scale. Their ecotoxicology, particularly to reef-building corals, is documented but incomplete: regulators and systematic reviewers have repeatedly noted that the dose-response data needed for marine risk assessment is missing for several compounds, including 4-methylbenzylidene camphor (4-MBC) [1, 2].

Against this background, 4-MBC is now the focus of accelerating international regulatory action. The European Union, the United Kingdom, China, and several ASEAN states have prohibited or are prohibiting the compound on the basis that no safe concentration can be established [3–5]. Vietnam has moved 4-MBC to its prohibited substances annex following the 42nd ASEAN Cosmetic Committee meeting [6]. The United States and Japan have never permitted it as a sunscreen active ingredient. Australia is now the only developed sunscreen market in its region in which 4-MBC remains permitted at concentrations of up to 4% w/w, and the only such market in which the national regulator has been formally asked to remove it and has declined to do so ([7]; IN-2025-AP-000920).

4-MBC, also known as enzacamene, is a camphor-based UV-B absorber. It is highly lipophilic, resistant to standard biological degradation, and stable under ultraviolet radiation [4]. These properties make it persistent in aquatic environments. It is not removed by conventional wastewater treatment [8], and it has been detected in coral tissue at tropical reef field sites under ambient conditions [2].

1.1 The international regulatory picture

International regulatory action on 4-MBC has accelerated sharply since 2020. Two reef-state jurisdictions moved first on environmental grounds: the Republic of Palau listed 4-MBC among 32 banned sunscreen ingredients under its International Coral Reef Protection Act [9], and Thailand prohibited 4-MBC and three other UV filters in all marine national parks from 4 August 2021 [10]. The compound is not listed as an approved active ingredient in the U.S. OTC sunscreen monograph (21 CFR Part 352), and FDA has issued a tentative determination that enzacamene is not generally recognized as safe and effective (GRASE) as a sunscreen active [11]. The compound is also absent from the exhaustive positive list of permitted UV absorbers under the Japanese Standards for Cosmetics [12], meaning it has never been permitted for cosmetic use in Japan.

The decisive scientific assessment came from the European Union in 2022. The Scientific Committee on Consumer Safety (SCCS), in opinion SCCS/1640/21, ‘*cannot conclude on the safety of 4-MBC*’ because of insufficient data on potential genotoxicity, and found that the use of 4-MBC at 4% in cosmetic products ‘*would not be safe*’ [4]. On the basis of that opinion, Commission Regulation (EU) 2024/996 moved 4-MBC from the permitted UV filter list (Annex VI) to the prohibited substances list (Annex II) of the EU Cosmetics Regulation, with prohibition effective across all 27 member states from 1 May 2025.

The United Kingdom reached the same conclusion independently. The Scientific Advisory Group on Chemical Safety (SAG-CS) found in 2025 that 4-MBC could not be considered safe at any concentration [3], and Statutory Instrument 2026/23 was laid before Parliament on 15 January 2026, prohibiting 4-MBC from 15 July 2026. China followed

within days: the National Medical Products Administration prohibited 4-MBC in cosmetic products under NMPA Announcement No. 6 of 12 January 2026 [13].

The ASEAN Cosmetic Committee voted at its 42nd meeting in November 2025 to move 4-MBC to Annex II (prohibited substances) of the ASEAN Cosmetic Directive. Vietnam has formally adopted the change through Official Letter 647/QLD-MP, with a national application date of 17 November 2028 [6]. Adoption by other ASEAN member states is rolling.

1.2 Australia's position is now an outlier

Against this international pattern, Australia stands alone. 4-MBC remains permitted at up to 4% in therapeutic sunscreen products as a grandfathered active ingredient that has never been subjected to a modern formal safety and efficacy assessment under current Therapeutic Goods Administration (TGA) standards. The compound remains available on Australian retail shelves and continues to enter Australian wastewater systems daily.

The TGA refused an application to remove 4-MBC from the Permissible Ingredients Determination (IN-2025-AP-000920). Applying the Australian Sunscreen Exposure Model adopted by the TGA in January 2025, the TGA evaluator established a systemic exposure dose for 4-MBC of approximately 1.4 mg/kg bw/day but was unable to derive a systemic No Observed Adverse Effect Level from the available data. The Delegate concluded that 'a MoS of 4-MBC cannot be reached, and consequently, a safe dose of the ingredient cannot be established'. The variation was nevertheless refused on the basis of 'the risk of sunscreen supply shortage, potentially leading to increasing rates of skin cancer', and the TGA's public-facing review of 4-MBC remains ongoing [7].

The central feature of the Australian situation is structural: the safety finding under the TGA's own exposure model has not been contested, but the regulatory action that would follow elsewhere has been declined on supply considerations rather than scientific ones.

The regulatory outlier status is the starting point for this Perspective, not its central question. The European, UK, Chinese, and ASEAN actions establish that 4-MBC cannot be considered safe under modern regulatory standards. Australia's continued permission of the compound at up to 4% in therapeutic sunscreens means that 4-MBC continues to enter the Australian environment at a rate determined by Australian sunscreen use, not by international regulatory consensus. The specific question this Perspective addresses is what happens to that 4-MBC load once it leaves the consumer: whether it reaches Great Barrier Reef (GBR) coastal waters, through what pathway, at what concentration, and with what biological consequence for already-stressed coral systems. Section 2 traces the wastewater pathway from population to ocean outfall. Section 3 reviews the experimental evidence on coral toxicity. Section 4 examines the specific Queensland reef zones where the discharge pathway and bleaching damage intersect.

1.3 Scope and sources

This is a narrative policy Perspective synthesising peer-reviewed literature, regulatory decisions, and Australian Government open data sources on 4-methylbenzylidene camphor (4-MBC) use, wastewater discharge pathways in Great Barrier Reef catchments, and coral toxicity. Peer-reviewed sources were identified through Google Scholar, PubMed using the search terms "4-MBC", "4-methylbenzylidene camphor", "enzacamene", "UV

filter”, “sunscreen”, “wastewater treatment”, “coral toxicity”, and “Great Barrier Reef”, and through forward and backward citation tracing from key reviews [1, 2, 32].

Regulatory sources were retrieved directly from the websites of the Therapeutic Goods Administration, the European Commission Scientific Committee on Consumer Safety, the United Kingdom Scientific Advisory Group on Chemical Safety, the National Medical Products Administration of China, the ASEAN Cosmetic Committee Secretariat, the United States Food and Drug Administration, and the Japanese Ministry of Health, Labour and Welfare. Australian wastewater infrastructure and discharge data were drawn from Queensland Government State of the Environment reporting, Townsville City Council and Cairns Regional Council publications, the Healthy Waters Partnership for the Dry Tropics Technical Report, and the Australian Institute of Marine Science Long-Term Monitoring Program.

Two Queensland reef zones and one downstream transport corridor were selected on the basis of documented coral condition data overlapping with confirmed wastewater treatment plant discharge points: the inshore reefs north of Cairns (Zone 1), Magnetic Island and Cleveland Bay (Zone 2), and the transport corridor from Townsville to the Palm Islands. Zone selection criteria were proximity of a major coastal wastewater treatment plant to documented reef monitoring sites, public availability of coral condition data, and population and tourism characteristics relevant to per-capita sunscreen load. Direct quantitative risk characterisation was not attempted; this is a narrative policy Perspective, and the absence of plant-specific 4-MBC influent and effluent data is itself part of the monitoring case made here.

2 The pathway: population, sunscreen load and direct ocean discharge

2.1 Population and sunscreen load in the two zones and transport corridor

Australian sunscreen use is materially higher than the European baselines that international 4-MBC exposure estimates have historically used. The Therapeutic Goods Administration’s Australian Sunscreen Exposure Model (ASEM), published in August 2025, was established specifically to reflect Australian conditions, which differ from European inputs in both UV climate and behavioural application patterns. The Australian Cancer Council recommends applying 140 mL of sunscreen for full-day full-body sun protection, a figure cited in the TGA’s own ASEM Impact Analysis [14]. European regulatory calculations use 16 to 18 g per day. The approximately eightfold difference reflects the higher UV index, longer outdoor season, and sun-exposure lifestyle of tropical coastal Australians, and it directly scales the volume of 4-MBC entering Queensland sewage systems relative to European-derived estimates.

Sunscreen applied to skin enters domestic sewage through showering, bathing, and pool use. Peer-reviewed literature identifies this indirect wash-off route as the primary pathway by which chemical UV filters reach wastewater treatment systems, because it concentrates the chemical load into a manageable treatment stream rather than dispersing it at source [8]. The critical variable is therefore how much sunscreen enters the system in the first place, which is a function of resident population, tourist population, and per-capita application rate.

The two Queensland reef zones and the transport corridor examined in this article are all served by high-use coastal centres in the tropical north, where year-round UV radiation requires near-daily sunscreen application. Townsville (serving both Zone 2 and

the Palm Islands transport corridor, encompassing the Cleveland Bay and Palm Islands receiving environments) has an Estimated Resident Population of 206,260 [15]. The city averages more than 300 sunny days per year, producing near-constant outdoor UV exposure and correspondingly elevated daily sunscreen use among residents. Townsville is also a regional tourism centre with access to the reef, Magnetic Island, and the Palm Islands group, although the resident population alone represents a sustained and predictable daily load into the Cleveland Bay Purification Plant and Condon Wastewater Treatment Plant (WWTP).

Cairns (Zone 1) has an Estimated Resident Population of approximately 180,000, but the resident figure substantially understates the actual sunscreen load. Tropical North Queensland received approximately 2.5 million domestic overnight visitors in the year ending March 2023 [16]. Great Barrier Reef Marine Park visitation reached 2.34 million visitor days in 2024, with approximately 86% of all GBR commercial tourism concentrated in the Cairns and Whitsunday planning areas [17]. Cairns alone accounts for the majority of this. On peak tourist days, visitor numbers entering the water for reef activities exceed 6,000 per day. Reef tourists are a particularly high-input category: they apply sunscreen repeatedly for full-day water exposure under intense UV conditions, then shower at tour-boat facilities and hotels, all of which feed into the Cairns wastewater network. The per-capita 4-MBC input to Cairns treatment plants is therefore likely to be materially above the national average used in the O'Malley et al. [8] study, both because tropical UV conditions drive higher resident usage and because concentrated high-application tourism activity is associated specifically with reef proximity.

2.2 Wastewater treatment does not fully remove 4-MBC

O'Malley et al. [8], in the only national-scale Australian study of UV filter release through wastewater treatment, sampled 33 wastewater treatment plants, with collection occurring predominantly during the August 2016 Australian Census period and a second campaign in December 2018. At least one UV filter was detected in 96% of effluent samples. The summed concentration of UV filters in effluent ranged from 130 to 8,400 ng/L, averaging 2,800 ($\pm$ 1,900) ng/L. Across the five UV filters quantified, the authors estimated that approximately 40% of the influent load passes through to effluent, with an estimated national release of approximately 20 kg per day of UV filters from treated wastewater into Australian aquatic environments [8].

For 4-MBC specifically, the average removal efficiency was 59 ($\pm$ 24)%, meaning approximately 41% of the influent 4-MBC load passes through to effluent unchanged. 4-MBC was detected in fewer than 20% of effluent samples in the national survey, with a maximum measured effluent concentration of 691 ng/L. The compound is not the most prevalent UV filter in Australian effluent (PBSA and BP4 dominate by detection frequency, at 95 and 58% detection respectively), but where 4-MBC is present in influent, the proportion reaching the environment is substantial [8].

The reason 4-MBC is not fully removed is its physicochemical profile. The compound is highly hydrophobic (log Kow approximately 5.1) and resistant to standard biological degradation processes used in conventional municipal treatment. Where 4-MBC is removed, removal occurs primarily through sorption to sludge during treatment rather than through biological breakdown [8, 18–20]. This creates a secondary pathway: where biosolids are subsequently applied to agricultural land, a practice common

across Queensland, 4-MBC and other hydrophobic UV filters can re-enter the broader environment via soil and runoff. The primary discharge route to coastal waters, however, remains direct liquid effluent through ocean outfalls and river discharge points.

Reliable removal of 4-MBC from wastewater requires advanced oxidation processes (photo-Fenton or UV/persulfate treatment), which have demonstrated 4-MBC degradation rates above 95% in laboratory and pilot studies [21, 22]. No Queensland municipal wastewater treatment plant currently employs advanced oxidation as part of its standard treatment train.

2.3 Effluent reaches coastal reef waters directly

The volume of treated sewage entering Great Barrier Reef catchment coastal waters is substantial. The Queensland State of the Environment Report 2020 documents combined annual discharge of approximately 290 gigalitres (GL) of treated sewage to coastal waterways across Queensland, of which approximately 230 GL originates from South East Queensland and the balance (approximately 60 GL per year) from Great Barrier Reef catchment coastal sewage treatment plants [23]. The two reef zones examined in this article are served by the most direct discharge pathways on the Queensland coast.

The Cleveland Bay Purification Plant, Townsville, services approximately 70 per cent of the city's population and operates at a post-2019 capacity of 87 (Mega Litres) ML per day following a major upgrade commissioned by Townsville City Council and completed in 2019 [24]. The plant discharges via an 800-metre ocean outfall within Cleveland Bay, Coral Sea. This is not a river or estuarine discharge: the pipe ends in open marine water within a partly enclosed coastal bay that adjoins the Great Barrier Reef Marine Park. Magnetic Island's reef systems lie approximately 20 km to the north-east, within the Marine Park boundary. Middle Reef, an inshore turbid-zone reef monitored under the AIMS Long-Term Monitoring Programme since 1993, lies approximately 4 km offshore from Townsville and 6 km from the city centre within Cleveland Bay itself [25, 26]. Condon WWTP, the second Townsville-region plant, discharges to the Bohle River, which enters Cleveland Bay from the north-west, contributing to the same semi-enclosed receiving environment as the Cleveland Bay outfall.

The Cairns wastewater network consists of six municipal treatment plants operated by Cairns Regional Council, with combined nominal capacity of approximately 54 ML per day [27]. The five plants with published capacity data, in order of throughput, are: Aeroglen (19.4 ML/day, servicing the Cairns central business district and northern beaches, discharging to the Barron River), Smithfield (8.3 ML/day, servicing the northern coastal suburbs from Yorkeys Knob to Palm Cove, discharging via a feeder drain to Half Moon Creek), Edmonton (6.7 ML/day, servicing southern Cairns suburbs, discharging to Trinity Inlet), Woree (servicing central and southern suburbs), and Gordonvale (1.9 ML/day, discharging to the Mulgrave River). All four receiving water bodies (Barron River, Half Moon Creek, Trinity Inlet, Mulgrave River) discharge to the Coral Sea within the Great Barrier Reef World Heritage Area. The inshore reef zone that recorded the steepest coral losses in the central Great Barrier Reef during the 2024 mass bleaching event lies immediately adjacent to these discharge points; Green Island reef lies approximately 27 km offshore of Trinity Inlet's mouth.

The inshore oceanography of the central and far-northern Great Barrier Reef coast carries this effluent toward reef systems rather than away from them. South-east trade

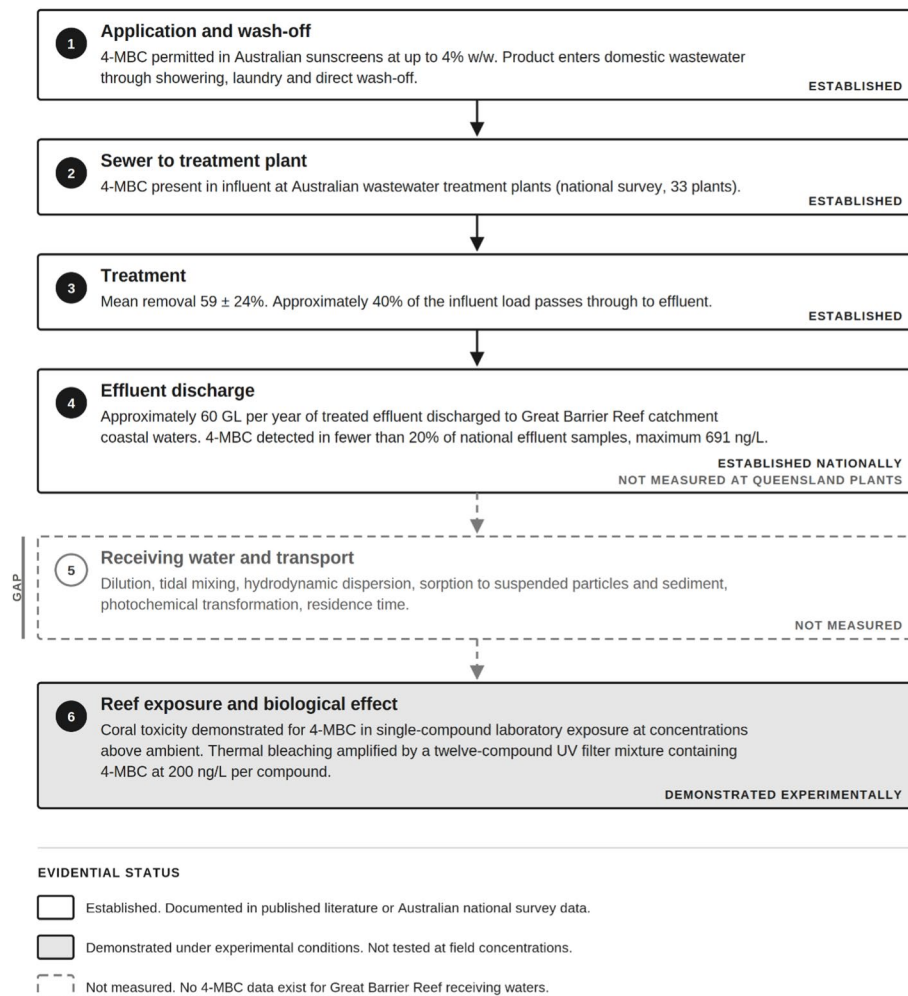


Fig. 1 Wastewater pathway for 4-MBC to Great Barrier Reef coastal waters, with the evidential status of each stage. Stages 1 to 4 are documented in the published literature and in Australian national survey data. Stage 5 has not been measured for 4-MBC at any Great Barrier Reef location. Stage 6 has been demonstrated under experimental exposure but not at field concentrations, which are unknown. The figure describes a pathway that is documented to the point of discharge and unmeasured beyond it. It does not represent measured environmental concentrations

winds dominate for approximately eight months of the year (April to November), driving net northward flow in inshore coastal waters. The eReefs hydrodynamic modelling system, developed jointly by CSIRO, the Australian Institute of Marine Science, the Bureau of Meteorology, and Queensland Government partners, provides three-dimensional simulation of river plume and particle transport across the Great Barrier Reef shelf at one-kilometre and four-kilometre resolution [26, 28]. Modelled and observational data indicate that coastal-introduced particles transit from inshore discharge points toward mid-shelf and outer-reef areas over periods of weeks to months, with transport pathways varying by season and discharge location. No prevailing current direction carries inshore effluent fully away from the Great Barrier Reef system. The pathway from permitted use to reef exposure, and the evidential status of each stage, is summarised in Fig. 1.

3 Biological effects: what 4-MBC does to coral

Direct laboratory exposure establishes 4-MBC as a coral toxicant. Danovaro et al. (2008) tested individual UV filters on *Acropora* species in field tanks at four reef sites (Caribbean, Celebes, Red, and Andaman Seas). 4-MBC, synthesised as a pure compound and dosed at the concentration equivalent to its 3% permitted use level in sunscreen, produced 63% chromatic-shift bleaching of *Acropora* sp. within 48 h at 33 $\mu\text{L/L}$; at 50 $\mu\text{L/L}$ it produced 95% bleaching of *Acropora pulchra* within 62 h, with 95% of zooxanthellae released. The mechanism is symbiont damage: loss of photosynthetic pigments, swollen and vacuolated zooxanthellae, and loss of cell-membrane integrity, with viral induction of latent infections in zooxanthellae proposed by the authors as the broader mechanism. The Danovaro 4-MBC exposure concentrations are higher than those measured in real coastal water; what the study establishes is the toxicant relationship and the cellular mechanism, not the field-relevant dose. The field-relevant dose evidence comes from two sources [29]. Lyu et al. [2] sampled coral tissue across nine coral species in the Luhuitou fringing reef, Hainan, and detected 4-MBC in 83.3% of coral tissue samples at concentrations of 0.0 to 2.2 ng/g dry weight. Total chemical UV filter burden in coral tissue across the eight compounds detected ranged from 4.3 to 52.0 ng/g dry weight, with bioaccumulation factors for 4-MBC higher in macroalgae than in coral. Lyu et al. specifically noted that coral toxicity data were sufficient to compute predicted-no-effect concentrations (PNEC) only for BP-1, BP-3, BP-8 and octocrylene; the available 4-MBC coral toxicity data were too limited to derive a PNEC, and 4-MBC was therefore not included in the study's quantitative ecological risk assessment [2]. The compound is consistently present in wild reef-building coral at a tropical tourist reef under ambient conditions; the dose-response data required to formally risk-assess it has not yet been generated. Tsui, Leung, [30], in a probabilistic risk assessment of surface waters from seven cities and the Arctic, found that 4-MBC posed greater risk to algae than to fish or coral [30]. The third category of evidence is the co-stressor experiment: what happens to coral exposed simultaneously to UV filters and thermal stress at concentrations that would be considered too low to matter in isolation. He et al. (2023) exposed three coral species (*Seriatopora caliendrum*, *Pocillopora acuta*, *Montipora aequituberculata*) to an equimolar mixture of 12 chemical UV filters including 4-MBC at 200 ng/L per compound (total mixture concentration approximately 2,400 ng/L), with and without thermal stress at 30 °C, over 60 days. Under thermal stress alone, the corals did not bleach severely. Under the UV filter mixture alone, 37.5% of *S. caliendrum* bleached and 12.5% died. Under combined exposure, 100% of *S. caliendrum* bleached with 100% mortality, 100% of *P. acuta* bleached with 50% mortality, and 25% of *M. aequituberculata* bleached with no observed mortality [31]. This finding sits in direct tension with the position taken in Wheate [32], which argued that UV filter effects on aquatic life occur only at concentrations 10 to 10,000-fold higher than those found in the environment and that observed reef damage is attributable to other causes [32]. He et al. (2023) demonstrate measurable, severe effect at 200 ng/L under realistic co-exposure with thermal stress; the Wheate position holds for single-compound acute laboratory endpoints but does not address the mixture-with-thermal-stress mechanism. The broader systematic review by Watkins and Sallach [1] catalogues the underlying problem on which both positions ultimately depend: current data are insufficient to fully determine the ecotoxicological risks of UV filter exposure in coral reef ecosystems, with co-stressor effects, long-term

exposure, and mixture toxicity identified as priority research gaps [1]. The Lyu et al. (2025) and He et al. (2023) findings begin closing that gap; the field-monitoring data required to apply either finding to a specific Australian reef does not exist.

The evidence on 4-MBC toxicity to coral spans three categories: controlled laboratory exposure studies establishing direct bleaching effects, field surveys documenting tissue uptake in wild coral, and co-stressor experiments testing UV filter mixtures against thermal stress. Each is relevant to the Great Barrier Reef context for a different reason. The first establishes that 4-MBC is a coral toxicant; the second establishes that the compound reaches and accumulates in reef-building coral under ambient conditions; the third establishes that the bleaching threshold of an already-stressed coral can be lowered by UV filter exposure at concentrations measured in real coastal waters.

4 Focus zones and transport corridor: bleaching damage and WWTP proximity

The following two reef zones and one downstream transport corridor have been selected on the basis of documented bleaching damage overlapping with confirmed WWTP discharge points. In each case, the primary cause of observed damage is identified before the UV filter co-stressor argument is assessed. The argument is not presented as an alternative explanation to climate change and known water quality threats; it is presented as an additional and unmonitored layer operating through the same coastal discharge infrastructure already identified as a problem.

4.1 Zone 1 — inshore reefs north of Cairns

Documented damage: In 2024, the Cairns–Cooktown sector recorded the largest single-year decline in hard coral cover in 39 years of Australian Institute of Marine Science Long-Term Monitoring Program (AIMS LTMP) data, with the Cooktown–Lizard Island sub-sector falling from a recorded high of 31.4% cover to 19.3% [33]. Cyclone Jasper made landfall north of Cairns in December 2023, delivering record rainfall and freshwater inundation across catchments draining onto inshore reefs [34]. The primary cause of the 2024 coral loss is unambiguous: a marine heatwave coinciding with Coral Sea summer sea surface temperatures that reached the highest level in at least 400 years, exceeding the previous record set in 2017 [35].

WWTP proximity and population context: Four treatment plants serving Cairns discharge approximately 54 ML/day in combined capacity into Trinity Inlet, Half Moon Creek and the Barron River, all of which open onto the Coral Sea within kilometres of the inshore reefs that recorded the worst losses [27]. The plants serve a resident population of approximately 180,000. In the year ending March 2023, Tropical North Queensland recorded approximately 13.8 million domestic visitor nights and \$4.3 billion in domestic overnight visitor expenditure [16]. A large proportion of these visitors participate in reef-based activities. Cairns reef tourists represent a high-input user category: they apply sunscreen repeatedly for full-day reef immersion in high-UV conditions, and the resulting wash-off enters the city's treatment system either directly through vessel-based showering and toilet facilities or indirectly through accommodation-based bathing. The O'Malley et al. [8] national average per-capita UV filter load is therefore likely to understate the load entering Cairns' plants relative to its population characteristics [8].

UV filter plausibility assessment: The corals that suffered the heaviest losses in the 2024 event were *Acropora*, fast-growing branching species with documented sensitivity

to thermal stress [33]. *Acropora* is also the genus in which direct 4-MBC bleaching was demonstrated by Danovaro et al. (2008), though at concentrations far above ambient [29]. The co-stressor evidence comes from a different set of species. He et al. (2023) exposed *Seriatopora caliendrum*, *Pocillopora acuta* and *Montipora aequituberculata* to an equimolar twelve-compound UV filter mixture at 200 ng/L per compound under 30 °C thermal stress. Bleaching occurred only under co-exposure; elevated temperature alone produced none. After 60 days, 100% of *S. caliendrum* bleached with 100% mortality and 100% of *P. acuta* bleached with 50% mortality, while *M. aequituberculata*, the only acroporid tested, showed 25% bleaching and no mortality [31]. He et al. did not test any *Acropora* species, and whether *Acropora* would respond like the pocilloporids or like the acroporid tested here was not addressed by that study.

The precautionary argument is testable: if inshore *Acropora* communities adjacent to these discharge points were receiving a continuous sub-lethal UV filter load from Cairns' treatment plants, their thermal bleaching threshold would be expected to be lower than in uncontaminated water. Whether sub-lethal UV filter exposure lowered the bleaching threshold of these communities cannot be confirmed or excluded without water quality monitoring, which has never been conducted.

4.2 Zone 2 — magnetic Island and Cleveland Bay

Documented damage. The Healthy Waters Partnership for the Dry Tropics 2023–2024 Technical Report, drawing on monitoring data collected from July 2023 to June 2024, grades Cleveland Bay inshore coral as 'poor' (D, score 37) against a Cleveland Bay inshore water quality grade of 'good' (B, score 76) in the same reporting period [36]. The report notes that Cleveland Bay coral scores and grades have fluctuated within this range for the past four years, which it attributes to several pressures including cyclones and increased water temperatures leading to bleaching, and identifies the Juvenile and macroalgae indicators as the primary drivers of the low scores, indicating limited coral recruitment and high macroalgal density competing for available substrate [36]. Of the seven coral reefs surveyed across Cleveland and Halifax Bays, four received a 'very poor' grade for macroalgae and all but one reef measured 'very poor' or 'poor' for juvenile coral counts [36]. Cyclones and thermal bleaching are documented pressures on Cleveland Bay coral. Macroalgal dominance and juvenile coral failure are documented in the inshore monitoring data. UV filter exposure has never been measured in the bay. The water-column monitoring framework grades water quality as good but does not include any contaminant in this chemical class. Whether UV filters are contributing to the cumulative pressure on Cleveland Bay coral cannot be established or ruled out from existing data. Macroalgal dominance and juvenile coral failure are the two indicators most directly associated with reef-community collapse, because they signal that the next generation of coral is failing to establish on substrate already taken by competing algae.

Middle Reef, a nearshore inshore-turbid-zone reef located approximately 4 km offshore from Townsville in Cleveland Bay, has been monitored by the Australian Institute of Marine Science Long-Term Monitoring Program since 1993 [25, 33]. The reef is a known sentinel site for the response of inshore corals to compounded stressors and shows a sustained trajectory of decline.

WWTP proximity and population context. The Cleveland Bay Purification Plant discharges through an 800-metre ocean outfall directly into open Cleveland Bay [37]. The

receiving water of that outfall is the same body of water in which Middle Reef sits, in which juvenile coral counts have been graded 'very poor' or 'poor' on six of seven monitored reefs across Cleveland and Halifax Bays, and in which macroalgae has dominated reef substrate at four of seven monitored reefs across the same two bays [36].

Cleveland Bay's physical geometry compounds this discharge geometry. The bay is a shallow, northward-facing embayment of approximately 325 km², landlocked on its southern and eastern sides by the mainland and bordered by Magnetic Island on its northern margin, with limited tidal flushing [25]. These are the same physical conditions identified in the international UV filter literature as producing the highest ambient concentrations of chemical UV filters in coastal water [38, 39].

The Cleveland Bay Purification Plant is already explicitly named within the Black Ross (Townsville) Water Quality Improvement Plan as the largest point-source nutrient emitter discharging to Cleveland Bay [40]. Biosolids produced by the same facility are used as fertiliser [41], establishing a re-entry pathway for solid-phase residuals back into the Reef catchment in addition to the direct ocean-outfall pathway for the liquid effluent.

UV filter plausibility assessment. Zone 2 is the most scientifically defensible site on the Queensland coast for the co-stressor hypothesis. Three factors converge.

1. *Discharge pathway directness.* The pathway from population to reef is the most direct on the Queensland coast: a single pipe, a single ocean-floor outfall, terminating in the bay where the reef decline is documented. There is no intermediate estuary, no river, and no offshore transport corridor between the outfall and Middle Reef. The plant has been identified as the largest WWTP point-source contributor in the catchment by the Queensland Government's own water quality improvement programme [40].
2. *Physical conditions favouring elevated ambient UV filter concentrations.* Cleveland Bay's semi-enclosed geometry, shallow water and limited tidal flushing are the bay characteristics associated with the highest ambient UV filter concentrations measured in Australian field studies. Carve et al. (2023), sampling 19 sites in and around Port Phillip Bay, reported total maximum chemical UV filter concentrations of 119 to 8,597 ng/L at high-recreation coastal sites against 74 to 108 ng/L at low-recreation sites. 4-MBC was the most widely detected filter, present at 9 sites and reaching a maximum of 1,643 ng/L, and the aquatic predicted-no-effect concentration for 4-MBC of 40 ng/L [42] was exceeded at those 9 sites [38]. A coral-specific PNEC for 4-MBC cannot currently be derived from available toxicity data [2], and the 40 ng/L value is used here as a general aquatic screening benchmark, not as a coral-relevant threshold. Allinson et al. (2018), in an earlier reconnaissance of four Port Phillip Bay estuaries, detected 16 of the screened UV filters and light stabilisers in water or sediment, including 4-MBC at 642 ng/L, but found that only two compounds, octocrylene and ethylhexyl methoxycinnamate, returned a risk quotient above 1, and concluded that few of the screened chemicals posed an individual short-term risk at the time of sampling [39]. Two features of these studies qualify their application here. Carve et al. attributed the concentrations they measured to direct recreational wash-off rather than to wastewater, and detected no UV filters at the river sites sampled on an ebbing tide, whereas Allinson et al. detected 4-MBC in the same bay. Both author groups call for further sampling in Port Phillip Bay. The two available Australian datasets therefore disagree on

whether the wastewater-influenced pathway delivers detectable UV filters to receiving waters, and both conclude that more measurement is required.

3. *Biological outcome matches predicted UV filter mechanism.* The third factor is that the observed biological outcomes match the predicted UV filter mechanism. Two biological outcomes have been documented in Cleveland Bay. Juvenile coral counts have been graded 'very poor' or 'poor' on six of seven monitored reefs. Macroalgal cover has been graded 'very poor' on four of seven monitored reefs across Cleveland and Halifax Bays. Both outcomes are consistent with the co-stressor mechanism demonstrated by He et al. (2023) for a UV filter mixture under thermal stress. That study was conducted in aquaria using a defined twelve-compound mixture, and its findings have not been tested against field conditions in Cleveland Bay or at any other Great Barrier Reef location. Nutrient-driven macroalgal competition provides an equally valid explanation for the same outcomes. The two mechanisms are not mutually exclusive. Neither can be confirmed nor excluded as a contributing factor to Cleveland Bay coral condition without water column and sediment monitoring for UV filters. No such monitoring has ever been conducted in the bay.

The same outfall delivers confirmed nutrient pollution into Cleveland Bay. This is already acknowledged by the Townsville City Council's own water quality improvement programme [40]. The same outfall is the unmonitored delivery point for any 4-MBC that enters the plant and survives treatment. Both the influent presence of 4-MBC in Australian wastewater [8] and its incomplete removal by conventional treatment processes [43] are established in the published literature. What has not been established at this plant is the concentration leaving the outfall, because it has never been measured. This is not a new pathway. It is an additional, unmonitored chemical through an acknowledged one. The case for monitoring is most difficult to dismiss here because it does not require establishing a new argument. It requires only extending an existing one.

4.3 Transport corridor: townsville to the palm Islands

The Palm Islands reef group sits approximately 65 km northwest of Townsville. Townsville is the largest local government area in regional Queensland, with an Estimated Resident Population of 206,260 at 30 June 2025 [15]. Its principal coastal treatment plants discharge into Cleveland Bay (Cleveland Bay Purification Plant via an 800 m ocean outfall) and into the Bohle River, which enters Cleveland Bay (Condon WWTP) [44]. Wind-driven surface circulation on the central Great Barrier Reef inner shelf is dominated by the south-easterly trade winds, which blow consistently through the winter dry season and drive northward-flowing coastal currents on the inner shelf [17]. Modelling studies of the central Great Barrier Reef confirm a vertical and cross-shelf disconnect between the wind-driven northward inner-shelf flow and the East Australian Current-driven southward mid-lagoon flow [45]. Devlin and Brodie [46], examining the dispersion of terrestrial discharge in the Great Barrier Reef lagoon, established that river plumes in the region are primarily dispersed northwards along the coast under wind and Coriolis effect [46]. The same physical regime carries treated effluent from the Townsville outfalls northward along the same coastal corridor, where it is joined by discharge from the Ingham and Innisfail treatment plants before reaching the latitude of the Palm Islands. No water quality monitoring programme has measured 4-MBC concentrations in this corridor at any point between the Townsville outfalls and the Palm Islands reefs, and the

present Perspective does not attribute observed Palm Islands reef condition to UV filter exposure. The corridor is identified here only to establish that the discharge geometry places Townsville WWTP effluent in continuous downstream proximity to a major reef group of the inshore Great Barrier Reef.

5 What this evidence base supports and what it does not

The authors do not claim that 4-MBC caused the 2024 Great Barrier Reef bleaching event. Climate-change-driven marine heatwave is the primary, sufficient cause. Coral Sea temperatures in 2024 were the warmest in 400 years [35]. The AIMS Long-Term Monitoring Program 2024–25 Annual Report attributes the largest spatial bleaching footprint ever recorded to climate-change-induced heat stress.

This Perspective claims the following.

1. Treated wastewater is discharged continuously into Great Barrier Reef catchment coastal waters at scale, providing a pathway through which 4-MBC enters reef-adjacent waters where present in influent. Approximately 60 gigalitres of treated sewage is released annually into Great Barrier Reef catchment coastal waters, representing approximately 20 per cent of the total volume released from Queensland sewage treatment plants [47]. The Cleveland Bay Purification Plant, servicing approximately 70 per cent of Townsville's population, operates at a post-2019 capacity of 87 ML per day and discharges through an 800-metre ocean outfall terminating in Cleveland Bay, where Middle Reef sits [24, 37, 48].
2. 4-MBC is a demonstrated coral toxicant under experimental exposure, and thermal bleaching has been amplified by a UV filter mixture containing it. Direct bleaching has been demonstrated in *Acropora* species under laboratory exposure [29]. Bioaccumulation in wild reef-building coral has been documented at 0.0 to 2.2 ng/g dry weight in 83.3% of sampled tissue across nine species [2]. Amplification of thermal bleaching under co-stressor conditions has been demonstrated at 200 ng/L [31]. The position taken in Wheate (2022), that UV filter concentrations are too low to cause environmental harm, holds for single-compound acute exposure but does not address the mixture-with-thermal-stress mechanism that He et al. (2023) demonstrate.
3. The discharge pathway has never been monitored. No Queensland authority has measured 4-MBC concentrations in Cleveland Bay, Trinity Inlet, Half Moon Creek, or any other GBR-adjacent coastal water body.
4. 4-MBC is prohibited, never permitted, or under enacted prohibition in the European Union, the United Kingdom, China, ASEAN states, the United States, and Japan. The compound remains legal in Australian therapeutic sunscreens at up to 4% w/w.
5. A contributing factor of this kind cannot be confirmed or excluded without monitoring. The absence of monitoring is not evidence of safety.

The monitoring rationale advanced here is not specific to chemical UV filters. No UV filter permitted in Australian sunscreens, whether chemical or mineral, has been subject to environmental monitoring in Great Barrier Reef coastal waters, and the sampling programme proposed below would screen for the principal permitted filters of both classes. A comparison of the environmental profiles of sunscreen product classes is outside the scope of this Perspective. The argument for action does not require establishing that 4-MBC caused observed bleaching. It requires only acknowledging that a coral toxicant,

banned by Australia's regulatory peers, reaches GBR catchment waters through a pathway already acknowledged by the Queensland Government as ecologically significant. That pathway has been left unmonitored under a regulatory architecture in which no single Australian authority is responsible for closing the loop.

6 What monitoring would resolve, and the regulatory route to closure

Resolving the question requires two stages, and they should not be conflated. The first is a reconnaissance survey capable of establishing whether 4-MBC occurs in reef-adjacent water. Targeted sampling of the Cleveland Bay outfall plume, the Townsville-region receiving waters, the four Cairns receiving waterways (Barron River, Half Moon Creek, Trinity Inlet, Mulgrave River), and inshore reef waters at Magnetic Island and the Palm Islands would establish within weeks whether 4-MBC is present and at what order of magnitude. Because the co-stressor effect demonstrated by He et al. (2023) arose from a mixture rather than from any single compound, 4-MBC concentrations cannot be interpreted in isolation, and the survey should screen the principal UV filters permitted in Australian sunscreens, chemical and mineral, alongside the target analyte. That is the threshold question, and for 4-MBC it is currently unanswered. The second stage is exposure characterisation, which a reconnaissance survey cannot deliver. Establishing whether 4-MBC concentrations are ecologically relevant would require repeated influent, effluent, receiving-water and sediment sampling, and potentially biota sampling, across seasons, tidal states and periods of differing sunscreen use, together with assessment of the processes governing the fate of 4-MBC between outfall and reef: dilution, tidal mixing, hydrodynamic dispersion, sorption to suspended particles and sediments, photochemical transformation and residence time. The first stage determines whether the second is warranted. The eReefs hydrodynamic modelling system, operated jointly by CSIRO, the Australian Institute of Marine Science, the Bureau of Meteorology, and Queensland Government partners, is already operational at one-kilometre resolution and can generate plume trajectory analysis from each outfall coordinate [26, 28]. No new infrastructure is required to complete the first stage.

The absence of monitoring is not unique to Australia, but Australia's position is. The European Union operates a Watch List mechanism under the Water Framework Directive requiring Member States to monitor listed substances in surface waters, and has applied it to chemical UV filters: ethylhexyl methoxycinnamate was listed in 2015 and removed in 2018 once the monitoring data supported a risk assessment, and oxybenzone, octocrylene and octisalate have subsequently been listed. 4-MBC was not among them. The European prohibition of 4-MBC proceeded on human health grounds and was not preceded by environmental monitoring of the compound. Palau, which prohibited 4-MBC in 2018 expressly to protect coral reefs, acted on a precautionary basis rather than on the strength of a monitoring programme. No jurisdiction, therefore, has monitored 4-MBC in receiving waters before acting on it. What distinguishes Australia is not that it has not monitored, but that it has neither monitored nor acted, while continuing to permit the compound at up to 4% w/w. The European Watch List provides a direct institutional model for the reconnaissance stage proposed here.

The prohibition recommended here rests on both bases, and they are cumulative rather than alternative. The human health basis is already complete. The SCCS, the SAG-CS and the TGA each examined 4-MBC and each concluded that a margin of

safety could not be established, and the European Union, the United Kingdom, China and ASEAN signatory states acted on that conclusion through binding prohibitions. No jurisdiction that has prohibited 4-MBC has subsequently reversed that decision. Australia's continued permission of the compound therefore does not rest on an affirmative finding that it is safe. It rests on a finding that removing it carries a supply risk. That is a market availability judgement, not a scientific one, and it is the only respect in which Australia's position differs from that of its regulatory peers.

The environmental basis is precautionary and is set out above. A documented discharge pathway into Great Barrier Reef catchment coastal waters, demonstrated coral toxicity under experimental exposure, and no measurement of 4-MBC at any point along that pathway. The absence of measurement is presently operating as a reason to continue permitting the compound. Under a precautionary standard it is a reason to restrict it pending measurement. The monitoring campaign proposed here is directed at the environmental basis; the regulatory instrument identified below is available on either basis independently.

The regulatory route is equally short. The TGA concluded in its January 2026 decision (IN-2025-AP-000920) that *"a MoS of 4-MBC cannot be reached, and consequently, a safe dose of the ingredient cannot be established"* [7]. The Delegate nonetheless declined to remove 4-MBC from the Permissible Ingredients Determination, citing *"the risk of sunscreen supply shortage, potentially leading to increasing rates of skin cancer"*. The European Union, the United Kingdom, China, and ASEAN signatory states reached the same scientific conclusion independently and acted on it through binding cosmetic ingredient prohibitions. The Therapeutic Goods (Permissible Ingredients) Determination under section 26BB of the Therapeutic Goods Act 1989 is the existing Australian regulatory instrument under which alignment with the international scientific consensus could be affected. Alignment would require no new primary legislation.

7 Conclusions

4-MBC remains legal in Australian therapeutic sunscreens at concentrations up to 4% w/w. It is prohibited or never permitted across the European Union, the United Kingdom, China, ASEAN signatory states, the United States, and Japan. In January 2026 the Therapeutic Goods Administration concluded that a safe dose of 4-MBC cannot be established ([7]; IN-2025-AP-000920). No corresponding restriction on Australian use has followed.

The evidence assembled in this Perspective supports three statements, which differ in evidential status.

- (a) Established. 4-MBC is present in Australian wastewater influent and is incompletely removed in treatment, and treated effluent is discharged continuously into Great Barrier Reef catchment coastal waters at scale. Whether 4-MBC is present in the effluent of the specific Queensland plants identified here has not been measured.
- (b) Demonstrated under experimental conditions. Coral toxicity has been shown for 4-MBC in single-compound laboratory exposure, and amplification of thermal bleaching has been shown for a UV filter mixture containing 4-MBC. Neither result has been tested at Great Barrier Reef field concentrations, which are unknown.
- (c) Established. The Queensland discharge pathway has never been monitored for this class of contaminant.

The authors do not claim 4-MBC caused the 2024 Great Barrier Reef bleaching event. Climate-change-driven marine heatwave is the primary, sufficient cause. The authors do claim that a coral toxicant, banned by Australia's regulatory peers, reaches reef-adjacent coastal waters through a pathway that has never been monitored, and that the absence of monitoring is not evidence of safety.

Two actions follow directly. First, a reconnaissance monitoring survey of the two identified Queensland zones and the transport corridor for 4-MBC and the principal co-occurring UV filters permitted in Australian sunscreens, using the operational eReefs hydrodynamic modelling system to interpret transport from outfall to reef, followed by exposure characterisation if the survey establishes occurrence. Second, amendment of the Therapeutic Goods (Permissible Ingredients) Determination under section 26BB of the Therapeutic Goods Act 1989 to align Australian regulation with the international scientific consensus on 4-MBC. Neither action requires new infrastructure, or new statutory authority. Both are within the capacity of existing Australian institutions to deliver.

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Author contributions

A.C. and J.M. jointly conceived the study and developed the regulatory argument. J.M. led the drafting of the manuscript; A.C. conducted the literature search, regulatory source analysis, and data extraction. Both authors reviewed and verified all primary sources and citations, critically revised the manuscript, approved the final version, and agree to be accountable for all aspects of the work.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethical approval

Not applicable. This Perspective did not involve human participants, human data, or animals, and no ethical approval was required.

Consent to participate

Not applicable. This Perspective did not involve human participants.

Consent to publish

Not applicable. This Perspective does not contain data from any individual person.

Competing interest

The authors are employed by VeganicSKN Australia, a manufacturer of mineral (non-nano zinc oxide) sunscreen products. J.M. is the Chief Executive Officer of VeganicSKN Australia. The chemical UV filters discussed in this article belong to a different product class from those the authors' employer manufactures, which the authors note as a potential competing interest. The article processing charge was paid by the Australian Sunscreen Council (ASC), a peak industry body of which VeganicSKN Australia is a partner organisation, as detailed in the funding statement. A.C. is also listed as an expert adviser by the ASC. The authors declare that these associations did not influence the study design, the analysis or interpretation of the literature, or the decision to submit for publication. This article addresses the regulatory framework applied to chemical UV filters and does not promote any specific product, brand, or manufacturer.

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